DAY 3 | PRE-READ

Blueprints, Good Habits and Your AI Pair Programmer

A light read before Day 3. About 20 minutes. Includes the OpenAI Codex setup steps and the do's and don'ts card.

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Why read this?

As programs grow, they need structure, the same way a growing company needs teams and roles. In the morning you learn object-oriented programming (OOP). In the afternoon you start working with **OpenAI Codex**, an AI coding agent inside VS Code. Please do the setup steps below before class.

1. Blueprints and buildings

An architect draws one blueprint and a builder uses it to build many houses. In programming the blueprint is a **class** and each house is an **object**.

Real world	Things it has (data)	Things it does (actions)
Support ticket	customer, issue, priority	assign, close
Bank account	owner, balance	deposit, withdraw
Online order	items, total, status	add item, pay, cancel

2. Families of things and planning for problems

A premium customer and a standard customer share most features but differ in a few rules. Describing the shared parts once is **inheritance**. A good restaurant has a plan for when an ingredient runs out; programs use **exception handling** to notice a problem and keep running.

3. Tidy toolboxes NEW (v4)

Every project needs its own set of libraries at known versions, like a labelled toolbox. A **virtual environment** keeps each project's tools separate. A **pinned requirements** file records exact versions. **pip-audit** checks those tools for known security problems.

4. Your AI pair programmer NEW (v4)

OpenAI Codex reads your code, explains it, writes new code and can run tests. It is fast but not always right. Think of it as a keen new colleague: helpful, but you review everything before it reaches a customer.

5. Codex setup checklist (do before class) NEW (v4)

- Install VS Code, Python 3.12 or newer and Git (see shared/Laptop_Setup_Guide.pdf).
- In VS Code, open Extensions and install **OpenAI Codex**.

- Have the program's OpenAI API key ready. Your trainer shares it securely. Paste it only into the Codex sign-in box when asked to sign in with an API key.
- Choose the approved default model. See docs/MODEL_GUIDE.md and docs/CODEX_GUIDE.md.
- Open a project folder (for example your Day 2 work) and ask Codex: "Explain this file."

6. Do's and don'ts card NEW (v4)

Do	Don't
Plan first, then small steps	Paste API keys, customer data or credentials
Include only the relevant files	Use full-access or bypass modes
Write or generate tests first	Ask to "improve everything"
Read every diff	Use a top-tier model for routine edits
Check your daily token usage	Merge AI code without tests and review

Key idea: the model proposes, the engineer decides.

7. Think about these before class

- Pick an app you use daily. What "things" does it manage, and what can each thing do?
- What could go wrong if you accepted AI-written code without reading it?
- Why might a project break when it uses a newer version of a library than yesterday?

Monozukuri (ものづくり): the craft of making things well